

Lamp Installation

Description

Install the following packages and host a sample dynamic website with help of database.

1. Install Apache Webserver
2. Install PHP
3. Install MySQL/MariaDB

Configuration files, extensions and versions should be noticed.

Deployment Prerequisites

Ensure that your target system has active root privileges (or sudo access), a properly registered subscription or configured repository manager (DNF), and an active network connection to download specific packages.

Step 1 — Install and Configure Apache Web Server

The Apache HTTP Server serves as the foundation for handling incoming client web requests. Execute the following sequence to provision and boot the server application:

1.1 Package Installation

```
yum install httpd -y
```

1.2 Service Management

Initialize and configure the Apache system daemon to execute persistence parameters across system restarts:

```
systemctl start httpd  
systemctl enable httpd
```

1.3 Verify Initialization Status

```
systemctl status httpd
```

1.4 Infrastructure Directory & Layout Structure

- **Default Document Root:** /var/www/html (The global dynamic context area)
- **Primary Configuration File:** /etc/httpd/conf/httpd.conf
- **Supplementary Module Configurations:** /etc/httpd/conf.d/
- **System Error and Output Logs:** /var/log/httpd/

1.5 Verify Deployment Version

```
httpd -v
```

To confirm active connectivity, navigate your browser to `http://SERVER-IP` to observe the default Apache welcome interface.

Step 2 — Install MariaDB Database Server

MariaDB provides the relational data storage tier for structural web application metrics.

2.1 Core Package Provisioning

```
yum install mariadb-server -y
```

2.2 Service Activation

```
systemctl start mariadb  
systemctl enable mariadb
```

2.3 Status and Verification

```
systemctl status mariadb  
mysql --version
```

Step 3 — Hardening & Securing MariaDB

Production environments necessitate runtime security posture compliance. Execute the underlying script to eliminate default test assets and restrict unauthorized authentication permissions:

```
mysql_secure_installation
```

During execution, configure parameters interactively following these enterprise hardening best practices:

- **Set root password? Y** (Define an internal complex alpha-numeric string)
- **Remove anonymous users? Y** (Disallows explicit unauthorized client mapping)
- **Disallow root login remotely? Y** (Restricts database administrator access parameters strictly to local context loops)
- **Remove test database? Y** (Purges insecure default schema visibility)
- **Reload privilege tables? Y** (Forces structural memory execution immediately)

Step 4 — Database Initialization & Account Management

Establish the persistence layout for your custom dynamic web platform:

4.1 Root Shell Login

```
mysql -u root -p
```

4.2 Relational Structure Creation

```
CREATE DATABASE sampled;

CREATE USER 'rejin'@'localhost' IDENTIFIED BY 'rejin123';

GRANT ALL PRIVILEGES ON sampled.* TO 'rejin'@'localhost';

FLUSH PRIVILEGES;

EXIT;
```

Step 5 — Install PHP and Associated Dependencies

PHP interprets server-side business logic and securely queries data pipelines to present structural interfaces.

5.1 Package Setup with Dependencies

```
dnf install php php-mysqlnd php-cli php-common php-gd php-mbstring php-xml php-  
json -y
```

5.2 Post-Install Module Validation

- **Core Engine Config:** /etc/php.ini
- **Version Output Check:** php -v

5.3 Engine Synchronization

Restart the HTTP Server infrastructure to trigger deep process links with the newly provisioned PHP compilation layers:

```
systemctl restart httpd
```

Step 6 — Create and Validate Sample Dynamic Website

Establish a real-time responsive functional check to verify internal communication pipelines between your Web Server layer and Database tier. This script includes explicit runtime error debugging configurations alongside an integrated PHP info report payload:

6.1 Target Document Construction

```
nano /var/www/html/index.php
```

6.2 Inject Unified Operational Codeblock

```
<?php

error_reporting(E_ALL);
ini_set('display_errors', 1);

$conn = new mysqli("localhost", "rejin", "rejin123", "sampledb");
```

```
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

echo "<h2>Database Connected Successfully</h2>";

echo "<hr>";

phpinfo();

?>
```

Save and close the document editor. Launch a browser window and process `http://SERVER-IP/index.php`. If the operational variables, user credentials, and database layouts match correctly, your system will securely report Database Connected Successfully followed by the foundational environment information page.

Technical Reference Appendices

Configuration File Mapping Table

Component / Environment Target	Global File Path Context Location
Apache Base Engine Configuration	/etc/httpd/conf/httpd.conf
Apache Modular Config Directory	/etc/httpd/conf.d/
PHP Operational Context Specification	/etc/php.ini
MariaDB/MySQL Native System Engine Options	/etc/my.cnf

Installed Modules & Extension Mapping

Module Name	Architectural Purpose Context
php-mysqlnd	Native Dynamic MySQL Driver communication management link.
php-cli	Command-line interface processing pipeline executor.
php-gd	Server-side image generation and graphic formatting wrapper.
php-mbstring	Multi-byte character management tool for non-ASCII variables.
php-xml	Extensible Markup Language extraction and query translation.

Firewall Configuration and Utility Commands

To safely pass HTTP requests through external firewalls, expose port boundaries explicitly:

```
firewall-cmd --permanent --add-service=http
firewall-cmd --reload
```

Output Pictures

Database Connected Successfully

PHP Version 8.3.29

System	Linux ip-172-31-42-97.ap-south-1.compute.internal 6.12.0-124.52.1.el10_1 x86_64 #1 SMP PREEMPT_DYNAMIC Sat Apr 11 20:17:08 EDT 2026 x86_64
Build Date	Dec 16 2025 14:32:42
Build System	Red Hat Enterprise Linux release 10.1 (Coughlin)
Build Provider	Red Hat, Inc.
Compiler	gcc (GCC) 14.3.1 20250617 (Red Hat 14.3.1-2)
Architecture	x86_64
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-ldap.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-enchant.ini, /etc/php.d/20-ffi.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gd.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-intl.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-mysqli.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-posix.ini, /etc/php.d/20-shmop.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-sysvmsg.ini, /etc/php.d/20-sysvsem.ini, /etc/php.d/20-sysvshm.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlreader.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-xsl.ini, /etc/php.d/30-mysql.ini, /etc/php.d/30-pdo_mysql.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmlreader.ini, /etc/php.d/40-zip.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API420230831.NTS
PHP Extension Build	API20230831.NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled

phpMyAdmin

Server: localhost | Database: my_database | Table: tbl_1

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	int(10)			No	None			Change Drop More
2	name	varchar(100)	latin1_swedish_ci		No	None			Change Drop More

☐ Check all ☐ With selected: [Browse](#) [Change](#) [Drop](#) [Primary](#) [Unique](#) [Index](#) [Spatial](#) [Fulltext](#)

[Print](#) [Propose table structure](#) [Move columns](#) [Normalize](#)

[Add](#) 1 column(s) after name [Go](#)

Indexes

No index defined!

Create an index on 1 column(s) [Go](#)

Partitions

No partitioning defined!

[Partition table](#)

[Information](#)

13.207.60.72/phpmyadmin/index.php?route=/sql Console Space usage Row statistics